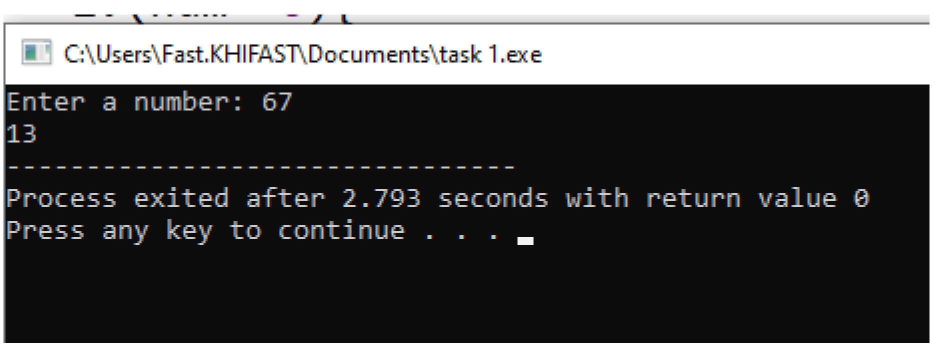
Task 1

```
#include<stdio.h>


int sum(int num){
	if(num==0){
		return 0;

  }
	return (num%10)+sum(num/10);
}


int main()
{
	int num;
	printf("Enter a number: ");
	scanf("%d",&num);
	printf("%d",sum(num));
	return 0;
}
```



Task 2

```
#include<stdio.h>
```

```
struct books{
	char title[50];
	char author[50];
	int page;
	float price;
};

int main()
{
	struct books book[3];
	for(int i=0;i<3;i++){
		printf("Enter title for book %d: ",i+1);
		fgets(book[i].title,50,stdin);
		printf("Enter author of book %d: ",i+1);
		fgets(book[i].author,50,stdin);
		printf("Enter pages for book %d: ",i+1);
		scanf("%d",&book[i].page);
		printf("Enter price for book %d: ",i+1);
		scanf("%f",&book[i].price);
		printf("\n");
		getchar();
	}

	for(int i=0;i<3;i++){
		printf("book %d:\n",i+1);
		printf("Title: %s\n",book[i].title);
		printf("Author: %s\n",book[i].author);
		printf("Pages: %d\n",book[i].page);
```

```
            printf("Price: %.2f\n",book[i].price);
            printf("\n");
        }

        return 0;
}
```

C:\Users\Fast.KHIFAST\Documents\Task 2.exe

```
Enter title for book 1: harry potter
Enter author of book 1: hassan
Enter pages for book 1: 123
Enter price for book 1: 34.98

Enter title for book 2: guardins
Enter author of book 2: hassan
Enter pages for book 2: 456
Enter price for book 2: 23.12

Enter title for book 3: hello world
Enter author of book 3: hassan
Enter pages for book 3: 789
Enter price for book 3: 87.90

book 1:
Title: harry potter

Author: hassan

Pages: 123
Price: 34.98

book 2:
Title: guardins

Author: hassan

Pages: 456
Price: 23.12

book 3:
Title: hello world

Author: hassan

Pages: 789
Price: 87.90


--------------------------------
Process exited after 51.23 seconds with return value 0
Press any key to continue . . .
```

Task 3

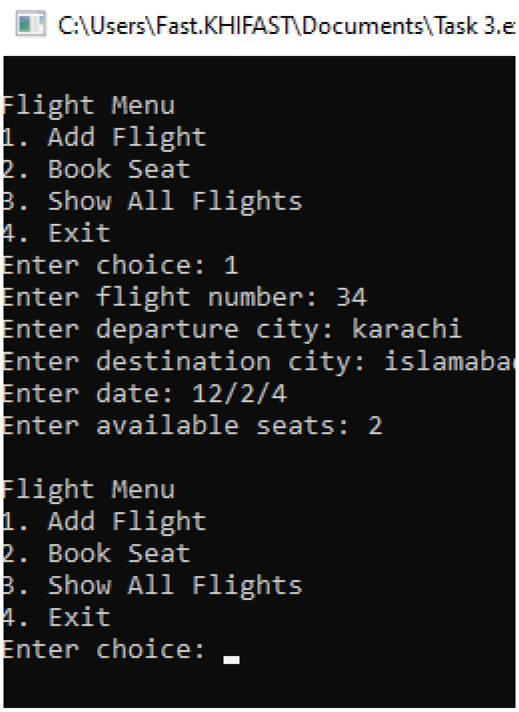
```
#include <stdio.h>
#include <string.h>

struct Flight
 {
   char flight_number[20];
   char departure[20];
   char destination[20];
   char date[20];
   int seats;
   static float price;
};

float Flight::price = 15000.0;

void bookSeat(struct Flight *f)
{
   if (f->seats > 0)
           {
      f->seats--;

      printf("Seat booked. Ticket price: %.2f\n", Flight::price);
   } else
           {
      printf("No seats available.\n");
   }
}
```

```
void showFlight(struct Flight f)
{
    printf("\nFlight Number: %s\n", f.flight_number);

    printf("Departure: %s\n", f.departure);

    printf("Destination: %s\n", f.destination);

    printf("Date: %s\n", f.date);

    printf("Seats: %d\n", f.seats);
}

int main()
 {
    struct Flight flights[5];
    int count = 0;
    int choice;
    int i;

    while (1)
            {
        printf("\nFlight Menu \n");

        printf("1. Add Flight\n");

        printf("2. Book Seat\n");
```

```
printf("3. Show All Flights\n");

printf("4. Exit\n");

printf("Enter choice: ");

scanf("%d", &choice);

if (choice == 1)
                {
   if (count < 5)
                        {
      printf("Enter flight number: ");
      scanf("%s", flights[count].flight_number);

      printf("Enter departure city: ");
      scanf("%s", flights[count].departure);

      printf("Enter destination city: ");
      scanf("%s", flights[count].destination);

      printf("Enter date: ");
      scanf("%s", flights[count].date);

      printf("Enter available seats: ");
      scanf("%d", &flights[count].seats);

      count++;
   } else
```

```
                                {
        printf("Flight list full.\n");
    }
}

else if (choice == 2)
                    {
    char num[20];
    printf("Enter flight number: ");
    scanf("%s", num);

    for (i = 0; i < count; i++) {
        if (strcmp(flights[i].flight_number, num) == 0);
                                            {
            bookSeat(&flights[i]);
        }
    }
}

else if (choice == 3) {
    for (i = 0; i < count; i++) {
        showFlight(flights[i]);
    }
}

else if (choice == 4) {
    break;
```

```
        }
    }

    return 0;
}
```


C:\Users\Fast.KHIFAST\Documents\Task 3.e
Flight Menu
1. Add Flight
2. Book Seat
3. Show All Flights
4. Exit
Enter choice: 1
Enter flight number: 34
Enter departure city: karachi
Enter destination city: islamaba
Enter date: 12/2/4
Enter available seats: 2
Flight Menu
1. Add Flight
2. Book Seat
3. Show All Flights
4. Exit
Enter choice:

Task 4

```
#include<stdio.h>


int main()

{

	int r;

	float Area;

	float const pi=3.142;

	printf("Enter the radius of the circle: ");

	scanf("%d",&r);

	Area=r*r*pi;

	printf("Area of circle: %.2f",Area);

	return 0;

}
```